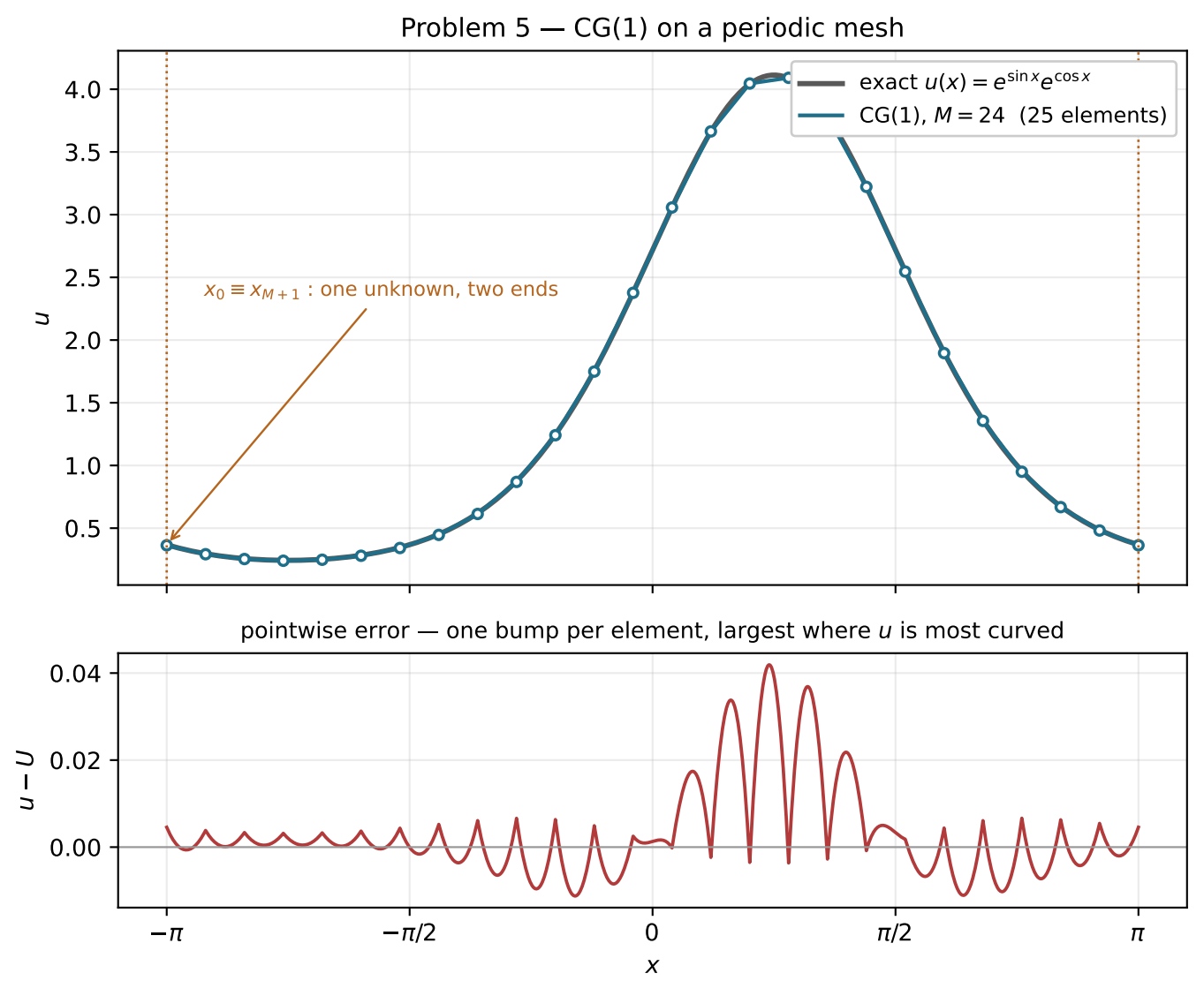
Problem 5 — CG(1) on a periodic mesh
exact $u(x) = e^{\sin x} e^{\cos x}$
CG(1), $M = 24$ (25 elements)
$x_0 \equiv x_{M+1}$: one unknown, two ends
u
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
pointwise error — one bump per element, largest where u is most curved
$u - U$
0.04
0.02
0.00
$-\pi$
$-\pi/2$
0
$\pi/2$
π
x